

IntegralAI

AI-Powered Integration Learning

Secure Authentication

IntegralAI

AI-Powered Integration Learning

Dashboard

Solver

Solutions

History & Progress

Practice

Elena Vance

Calculus II

Computational Access Portal

Rigorous symbolic, derivative, step-by-step calculus, and higher-order algebra workspaces.

Sign In

OAuth Account

STANDARD CREDENTIALS

Mathematical / Privacy Policy

secure@integralai.edu

Secure Token / Session

Logout

Remember this workspace (30 days)

OK

Sign Into Integrals

• Enhance the workspace with AI

Use proof verification-enabled integrations (check for domain consistency in real time)

Welcome back, Elena

Calculus II (MATH 202) Honors Track • Department of Mathematics

Calculus II Mastery Matrix

Power Rule & Polynomials

42 Solved • 0 Missions

Returned

91%

U-Substitution

28 Solved • 2 Incomplete

Proficient

84%

Integration by Parts

10 Solved • 5 Tutorial gaps

Needs Practice

62%

Trigonometric Substitution

17 Solved • 1 Triangle Mapping

In Progress

45%

Partial Fraction Decomposition

Production Equations • 4 Solved

New Topic

23%

Recent Analytical Derivations

$\int x \cdot e^{-(2x)} dx$

By Parts (Library) • Solved 32 min ago

4 Steps

Review

$\int (2x + 5) / (x^2 + 5x - 6) dx$

Logarithmic (Library) • Solved 3 hrs ago

3 Steps

Review

$\int \sqrt{9 - x^2} dx$

Trig Sub (Library) • Yesterday

Assisted

Review

$\int \sec(3x) dx$

Cyclic Parts (Revision) • 2 days ago

4 Steps

Review

Weekly Solver Volume

54 Integrals

94.2%

Quick Formula Sheets

PDF • CAS

Trigonometric Integrals & Identities

Partial Fraction Decomposition Rules

© 2024 IntegralAI Platform. Rigorous higher-level calculus & analytical reasoning tools.

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Calculus II

Curriculum

Calculus II (MATH-202)

Engineer: Sympy 1.12 CAS Active

Latency: 4ms

Solve an Integration Problem

Enter your integration expression or scan handwritten equations for step-by-step resolution.

Mathematical Expression Input

x^2

dx

$x \in \mathbb{R}$

INSERT

x^2

x

e^x

$\sin$

$\cos$

$\ln$

a/b

$\ln(a-b)$

Clear

Solve Integration

Learn. Solve. Understand.

Get step-by-step solutions for your integration problems with complete algebraic proofs and rigorous calculus explanations.

SAMPLE PROBLEMS

$\int x^2 dx$

$\int e^{(2x)} dx$

$\int \sin(x)\cos(x) dx$

$\int x \cdot \ln(x) dx$

Standard Table of Integrals

48 Fundamental Antiderivative Rules

Step-by-Step Solution

Method: Power Rule

Copy LaTeX

Export PDF

Bookmark

Problem: $\int x^2 dx$

Continuous on $\mathbb{R}$

Order: 1

Step 01: Identify the applicable integration rule

The integrand x^2 is of the polynomial power form ax^n where exponent $n = 2$ (and $n \neq -1$). Therefore, the fundamental Power Rule of integration applies.

Standard Form

$\int x^n dx = \frac{x^{n+1}}{n+1} + C$

Step 02: Apply the power rule

Increment the current exponent by 1 and divide the whole term by the resulting numerical exponent: $2 + 1 = 3$.

Substitution

$\frac{x^{2+1}}{2+1} + C$

Step 03: Simplify the expression

Perform simple integer arithmetic in the exponent and the denominator.

Arithmetic

$\frac{x^3}{3} + C$

Step 04: Add the constant of integration

Since this is an indefinite integral with unbounded domain intervals, include the arbitrary real integration constant $C \in \mathbb{R}$.

Constant Property

FINAL ANSWER

Antiderivative family of the polynomial integrand $f(x) = x^2$.

$\frac{x^3}{3} + C$

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Calculus II

Performance & Derivation Archive

Complete mathematical audit logs, verification trajectories, and algorithmic proficiency across real-variable techniques.

Solved History 68

Analytics & Mastery

Search integrals by formula (e.g., $\int \ln(x) \cdot e^{(x)} dx$), method, or tag...

OK

100% Sympy Verified

ALL METHODS

#048

CALC II

Oct 24, 2024 • 14:22 EST • Integration by Parts • LATE Rule

$\int x^2 e^{3x} dx = \frac{1}{3} x^2 e^{3x} - \frac{2}{9} x e^{3x} + \frac{2}{27} e^{3x} + C$

Verified

4 Steps

View Proof

#045

CALC II

Oct 20, 2024 • 17:48 EST • Trig Integrals • Half-Angle Reduction

$\int \sin^4(x) dx = \frac{3}{8} x - \frac{1}{4} \sin(2x) + \frac{1}{32} \sin(4x) + C$

Verified

5 Steps

View Proof

1

2

3

Next